# Kaixiang **Mai**

Computer Vision

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## Education

**Guangdong University of Technology** — *Guangzhou, China*

B.S. Candidate in Computer Science — *Sept. 2023 - Present*

- School of Computer Science and Technology
- Expected graduation: 2027

## Research Experience

**3D Medical Segmentation** — *MVP Research Group*

Undergraduate Researcher, advised by Prof. Yuzhu Ji — *2025 - Present*

- Investigated efficient and structure-aware representation learning for **3D medical image segmentation**, with a focus on capturing long-range dependencies while preserving complex anatomical structures.
- Developed **OcTreeMamba**, introducing anatomy-adaptive Octree representations with hierarchical state-space propagation to model sparse 3D anatomical structures while preserving structural continuity and reducing redundant background modeling. The work was **accepted to IEEE BIBM 2026**.
- Extended this line of research to **AdaOctNet**, treating the Octree as a task-driven semantic representation with spatially adaptive granularity learned end-to-end from volumetric features.
- Evaluated on 3D medical image segmentation benchmarks, including vascular and large organ datasets. Benchmarked against CNN-, Transformer-, and SSM-based architectures.
- Led problem formulation, model development, implementation, experimental evaluation, ablation studies, and manuscript preparation as the **first author**.
- Through these projects, my research focus evolved from improving a specific medical image segmentation task to studying how sparse, hierarchical, and topology-aware representations can encode complex 3D structures more effectively, shaping my broader interest in **representation learning for 3D vision**.

**Automatic Facial Pain Grading from Multi-view Facial Information** — *Undergraduate Research Project*

Student First Author; Co-inventor — *2024*

- Developed a computer vision-based approach for automatic facial pain grading using multi-view facial information.
- Contributed to method design, system formulation, and patent preparation.
- Invention patent application: “Facial Pain Grading Method, System, Device, Medium, and Product.”

## Publications

International Conference

1. **Kaixiang Mai**, Jiaqi Yu, Yuzhu Ji, Yiqun Zhang, Haijun Zhang, and Yiu-ming Cheung. “OcTreeMamba: Topology-Aware Mamba for 3D Medical Segmentation via Octree Sparsification”, accepted at the **2026 IEEE International Conference on Bioinformatics and Biomedicine (BIBM 2026)**, 2026.
2. **Kaixiang Mai**, Mingshan Sun, Peng Liu, Yuzhu Ji, Yiqun Zhang, and Yiu-ming Cheung. “AdaOctNet: Learning Anatomy-Adaptive Octree Representations for 3D Medical Image Segmentation”, Manuscript submitted to AAAI 2027, 2026.

## Academic Services

2026 **Reviewer,** IEEE Transactions on Emerging Topics in Computational Intelligence (TETCI)